U-S-350a

APRIL 16, 1957

SUPERSEDING
Int. Fed. Spec. U–S–00350 (GSA–FSS)
July 26, 1956, and
Fed. Spec. U–A–451b
September 13, 1946

FEDERAL SPECIFICATION

SILVER ALLOY POWDERS, DENTAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers silver alloy powders for dental amalgam suitable for use in the restoration of lost tooth structure.

1.2 Classification.

1.2.1 *Types and sizes.*—Silver alloy powders for dental amalgam shall be of the following types and sizes, as specified (see 6.2).

Type I.—Filings.

Type II.—Shavings.

Size 1. — 1 ounce troy (minimum net weight).

Size 2. — 2 ounce troy (minimum net weight).

Size 3. — 5 ounce troy (minimum net weight).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).

PPP–B–566 — Boxes, Folding, Paperboard.

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

PPP–B–676 — Boxes, Set-Up, Paperboard.

Federal Standard:

No. 102—Preservation, Packaging and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–L–10547 — Liners, Case, Waterproof.

Military Standard:

MIL–STD–129 — Marking for Shipment and Storage.

402173/57/No. 318

FSC Class 6520

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following publication of the issue in effect on date of invitation for bids, forms a part of this specification:

American Dental Association Specification:
No. 1 for Dental Amalgam Alloy.

(Copies of American Dental Association specifications may be obtained free upon application to the American Dental Research Fellowship, National Bureau of Standards, Washington 25, D. C.).

3. REQUIREMENTS

3.1 Comminuted alloy.—The comminuted alloy shall be free from foreign materials and shall be uniform.

3.2 Blackening.—When rubbed against the hand or paper, the amalgamated alloy shall not produce excessive blackening of the hand or white paper.

3.3 Working qualities. — Silver alloy for dental amalgam shall possess the following satisfactory working qualities:

3.3.1 *Time of amalgamation.*—The amalgam shall be ready for condensation not longer than 3 minutes after the trituration is initiated.

3.3.2 *Consistency.*—The amalgam shall not have a granular or sandy consistency when amalgamated.

3.3.3 *Carving.*—Using a Frahm carver, the amalgam shall remain susceptible to carving for at least 15 minutes after amalgamation, and shall no longer be carvable after 20 minutes from the start of amalgamation.

3.3.4 *Retention of polish.*—The amalgam shall receive and retain a polish 24 hours after amalgamation.

3.3.5 *Plastic amalgamation.*—Unless otherwise specified, a smooth plastic amalgamation shall be obtained in less than 1½ minutes.

3.4 Chemical composition.—The chemical composition shall be within the following limits:

Silver	68 percent,	minimum.
Tin	25 percent,	minimum.
Copper	5 percent,	maximum.
Zinc	2 percent,	maximum.

In addition to the above, the alloy may contain only gold and platinum.

3.5 Properties after amalgamation.—After the alloy is amalgamated and condensed in accordance with the instructions recommended by the manufacturer, the resulting amalgam shall comply with the following requirements:

3.5.1 *Flow.* — Twenty-four hours after amalgamation is started, when tested as described in 4.3.1, the average flow of the specimens shall not exceed 4 percent.

3.5.2 *Setting changes.*—Twenty-four hours after amalgamation is started, when tested as described in 4.3.2, the average length of the specimens shall have increased between 3 and 13 microns per centimeter.

3.6 Instructions.—Accurate and adequate instructions for proportioning and manipulation shall accompany each package. The following details shall be included in these directions: Ratio of alloy to mercury, type of mortar and pestle, force to be applied to pestle, mixing time, mixing speed (revolutions of pestle), when and how the excess mercury shall be expressed, and the method of packing and polishing.

3.7 Certification.—Silver alloy for amalgam shall have been tested at the National Bureau of Standards for the purpose of establishing a list of certified materials. Bidders shall attach to their bid a certificate and data show-

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ing that the brand of material they propose to furnish is the same in quality and brand as that appearing on the list of certified materials.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Sampling.—At least one container shall be taken at random from every 500 ounces or fraction thereof of each type for test at time of delivery.

4.2 Inspection.—Samples shall be inspected to determine compliance with applicable requirements of this specification.

4.3 Tests.

4.3.1 *Flow.*—The specimen shall be a cylinder 4 millimeters in diameter and 8 millimeters long, which shall be prepared by condensing the amalgam into a rigid steel die having a cavity 4 millimeters in diameter and approximately 11 millimeters long, using as far as possible the technique given in the sheet of instructions accompanying the alloy. The specimen shall be removed from the die on completion of condensation. Prior to insertion in the flow micrometer, the ends shall be surfaced plane and at right angles to the axis so that the length of the specimen is 8 millimeters. Three hours after amalgamation is started, the specimen shall be subjected to a constant axial load of 250 kilograms per square centimeter (approximately 3,550 lbs. per square inch). This load shall be maintained 21 hours. The change in length of the specimen shall be determined for compliance with the requirements of 3.5.1. During this test the temperature of the specimen shall be maintained between 20° and 25° C. The average shortening obtained from two such specimens shall be the recorded value for flow.

4.3.2 *Setting changes.*—The specimen shall be prepared by condensing the amalgam into a rigid steel die having a cavity approximately 5 millimeters in diameter and 10 millimeters long, using as far as possible the technique given in the sheet of instructions accompanying the alloy. The specimen shall be removed from the die as soon as condensation is completed and shall not be subjected to restraint during the test. The specimen shall be placed in the measuring instrument and adjusted within 12 minutes after amalgamation is started. Measurements shall begin 15 minutes after amalgamation is started. Measurements shall be made to determine compliance with the requirements of 3.5.2. During this test the temperature of the specimen shall be maintained between 20° and 25° C. The average change in length of two such specimens shall be the recorded value for setting change.

4.3.3 *Working qualities.* — Practical tests shall be made to determine working qualities after amalgamation.

5. PREPARATION FOR DELIVERY

5.1 Packaging. (See Federal Standard 102.)

5.1.1 *Unit of issue.* — One, two, or five ounces of silver alloy powder, as appropriate for the stock number specified in the invitation for bids, constitutes one unit of issue.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table 1 shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE I.—*Packaging quantities*

Federal Stock Number	Unit quantity	Unit package	Intermediate package	Shipping container
	(*oz.*)			
6520–500–2050 ----------	5	1 unit	6 units	96 units

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5.1.2.1 *Packing variation permitted.*—If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit packaging.*—Each unit shall be packaged in a glass bottle of suitable size and design, fitted with either a screwcap closure or a polyethylene snap cap secured with an aluminum rolled on closure. Each bottle (1 unit) shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Each carton shall be adequately secured to prevent accidental opening.

5.1.3.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification LLL–B–631 or PPP–B–676. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing. (See Federal Standard 102.)

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table IV.

5.2.1.2 *Case liner.*—Each box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table II.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Bottles.* — Each bottle shall be marked with stock number, item identification, name of contractor, lot (batch) number, date of manufacture (year and month), and the legend "Guaranteed to meet ADA Specification No. 1."

5.3.2 *Unit cartons.*—Each unit carton shall be marked with stock number, item identification, unit, quantity, name of contractor, lot (batch) number, date of manufacture (year and month), office identifying number, and the legend "Guaranteed to meet ADA Specification No. 1."

5.3.3 *Intermediate cartons and shipping containers.*—Each intermediate carton and each shipping container shall be marked in accordance with Military Standard MIL–

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STD–129. Lot (batch) number shall be shown. Date of manufacture shall be shown in lieu of date packed. Service color marking and symbol are not required on containers for level A packs when addressed to domestic establishments for the Medical Services of the Armed Forces.

6. NOTES

6.1 Federal Standard 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 Purchasers should exercise any desired options offered herein and specify unit quantity required. (See 1.2, 5.1.1, 5.1.2, and 5.1.3).

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the type, etc., normally procured to meet Federal Government requirements.

6.5 Transportation description.

6.5.1 Classified as precious metals, not taken by freight. Ship via Parcel Post or Express.

6.6 This specification covers the following item appearing in the Armed Services Medical Stock List:

6520–500–2050 — SILVER ALLOY POWDER, DENTAL 5 oz.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force—AF.

Copies of this specification may be purchased for 10 cents each.

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